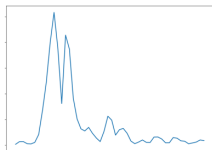


CIR Processing & Localization Running on Smartphone

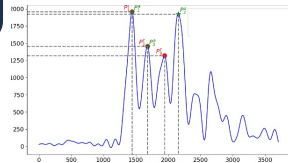
IMU-based Cross-Seat
Movement Detection



CIR Acquisition from
UWB ranging



Peak Identification and
Tracking



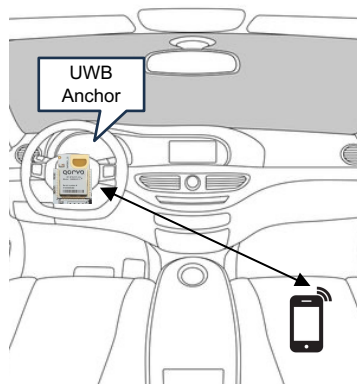
Peak-based CIR
Feature Extraction

Lightweight ML
Classifier

Driver or
Passenger Seat?

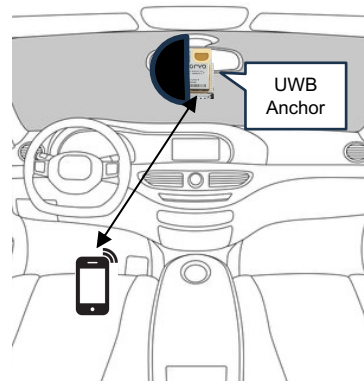
UWB Anchor Setup

Steering Wheel Column Mount



Asymmetric UWB anchor
placement

Rearview Mirror Mount
w/ half-cylindrical metal shield



Shielded UWB anchor